strips are absent in the endodermis cells which are located opposite the protoxylem elements. These thin-walled cells without casparian strips are called **passage cells** through which water and mineral salts are conducted from the cortex to the xylem elements. Water cannot pass through other endodermal cells due to the presence of casparian thickenings.

Stele

All the tissues present inside endodermis comprise the stele. It includes pericycle and vascular system.

Pericycle

Pericycle is generally a single layer of parenchymatous cells found inner to the endodermis. It is the outermost layer of the stele. Lateral roots originate from the pericycle. Thus, the lateral roots are endogenous in origin.

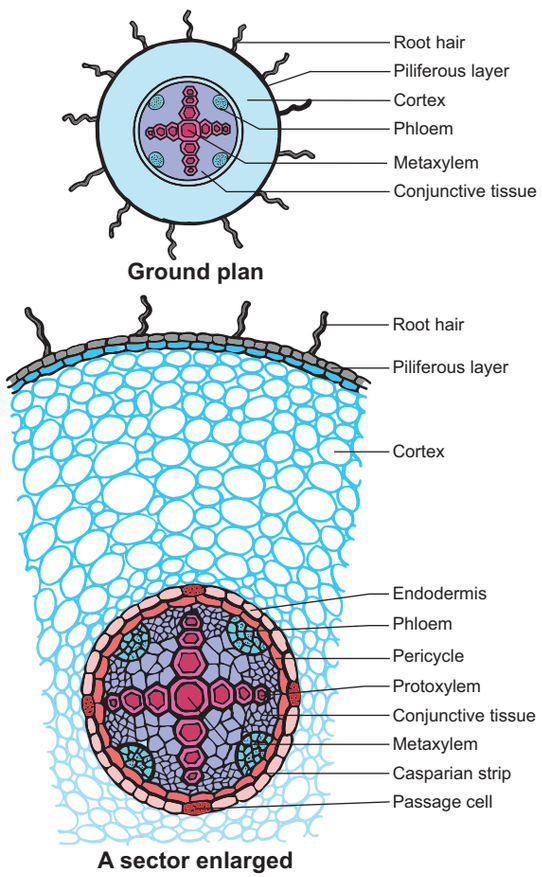


Figure 9.16: T.S. of Dicot root (Bean root)

Vascular System

Vascular tissues are in **radial arrangement**. The tissue by which xylem and phloem are separated is called **conjunctive tissue**. In bean, the conjuctive tissue is composed of parenchyma tissue. Xylem is in **exarch condition**. The number of protoxylem points is four and so the xylem is called **tetrach**. Each phloem patch consists of sieve tubes, companion cells and phloem parenchyma. Metaxylem vessels are generally polygonal in shape. But in monocot roots they are circular.

Primary Structure of Monocot

Root-maize Root

The transverse section of the monocot root (maize) shows the following plan of arrangement of tissues from the periphery to the centre.

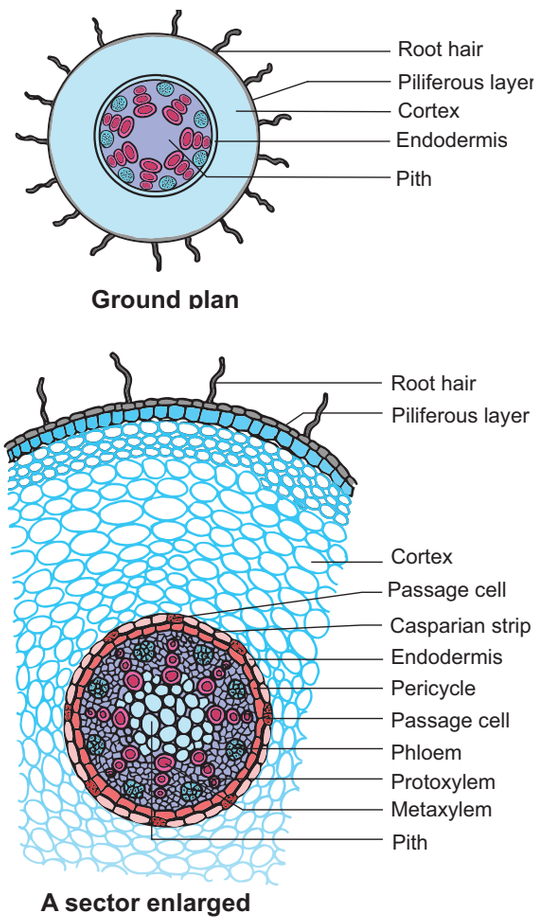


Figure 9.17: T.S of Monocot root (Maize root)

Piliferous Layer or Epiblema

The outermost layer of the root is known as **piliferous layer**. It consists of a single row of thin-walled parenchymatous cells without

XIth_Botany_Chapter_9_EM.indd 210 5/17/2020 9:16:39 PM